Supplementary Note 3

Details of Dimensional Reduction Analysis

Supplementary Note Discussion SND 3-1

Dimensional Reduction Analysis

PCA[1] is a powerful dimensionality reduction technique that has been widely adopted across various domains, from data science and machine learning to computer vision and bioinformatics. At its core, PCA aims to transform high-dimensional data into a lower-dimensional space while preserving the most important information and patterns within the data. The underlying principle of PCA is to identify the directions of maximum variance within the data, known as the principal components. These principal components are essentially the eigenvectors of the covariance matrix of the data, sorted in descending order of their corresponding eigenvalues. By projecting the data onto the subspace spanned by the top principal components, PCA can effectively capture the most significant sources of variation, effectively reducing the dimensionality of the data without compromising the essential characteristics. One of the key advantages of PCA is its ability to remove noise and redundant features from the data, making it a crucial preprocessing step for many machine learning algorithms. By reducing the dimensionality of the data, PCA can also improve the computational efficiency of downstream analyses, as well as facilitate better data visualization and interpretation. Additionally, PCA can be used to identify the most influential features or variables within a dataset, which can provide valuable insights for domain experts and decision-makers.

In contrast, t-SNE[2] is a nonlinear dimensionality reduction technique that excels at visualizing high-dimensional data in a low-dimensional space, typically two or three dimensions. Unlike PCA, which focuses on preserving the global structure of the data, t-SNE is primarily concerned with preserving the local structure, ensuring that similar data points in the high-dimensional space are represented by nearby points in the low-dimensional embedding. The t-SNE algorithm works by first computing the pairwise similarity between data points in the high-dimensional space, using a Gaussian kernel to model the probability distribution. It then seeks to find a low-dimensional representation of the data where the pairwise similarities are preserved as accurately as possible, using a Student's t-distribution to model the similarity in the low-dimensional space. By minimizing the Kullback-Leibler divergence between these

two probability distributions, t-SNE is able to generate a faithful low-dimensional projection of the data, often revealing intricate structures and patterns that might not be easily discernible in the original high-dimensional space. The power of t-SNE lies in its ability to capture both global and local structures within the data, making it particularly useful for exploratory data analysis, cluster identification, and data visualization. It has been widely adopted in various fields, including bioinformatics, image processing, and natural language processing, where the visualization of high-dimensional data can provide valuable insights and aid in the understanding of complex relationships within the data.

Overall, PCA and t-SNE are complementary dimensionality reduction techniques that play crucial roles in the field of data science and machine learning, each with its own unique strengths and applications. By leveraging these powerful tools, researchers and practitioners can gain a deeper understanding of their data, uncover hidden patterns, and ultimately make more informed decisions.

Building upon our preliminary unsupervised data mining results, we employed these techniques to further examine our dataset. This examination revealed a significant overlap between positive and negative samples across various thresholds identified in our analysis by dimensional reduction of our domain knowledge dataset from high dimension to two-dimensional space as shown in **Figure SN 3-1-Figure SN 3-2**. Such an overlap is indicative of the complexity and nuanced nature of our dataset, suggesting that simpler ML methods might fall short in providing accurate and detailed predictions. This is particularly evident in the Silhouette Coefficients and Davies-Bouldin Index values obtained from our analysis (**Table SN 3-1**). The Silhouette Coefficients were supposed to be values close to +1, but were found notably low by the PCA and t-SNE results, not exceeding +0.2 and occasionally entering negative territory, indicating poorly defined clusters with significant overlap. Meanwhile, the Davies-Bouldin Index values were notably high, as they were supposed to be values less than +1 and closer to 0. This has further confirmed the lack of distinct, well-separated clusters in our data. These metrics underscore the challenges in analyzing and interpreting our dataset, which is characterized by high-dimensional complexity and indistinct separability of samples. Therefore, these findings underscore the necessity of employing more sophisticated ML strategies in our subsequent analysis. Advanced ML approaches are essential to navigate

the intricacies of our multifaceted dataset effectively, enabling us to discern the underlying patterns and relationships with greater precision and nuance.

The scripts of this part are stored in the GitHub repository in the directory “/PCA and t-SNE Dimensional Reduction Representation”.

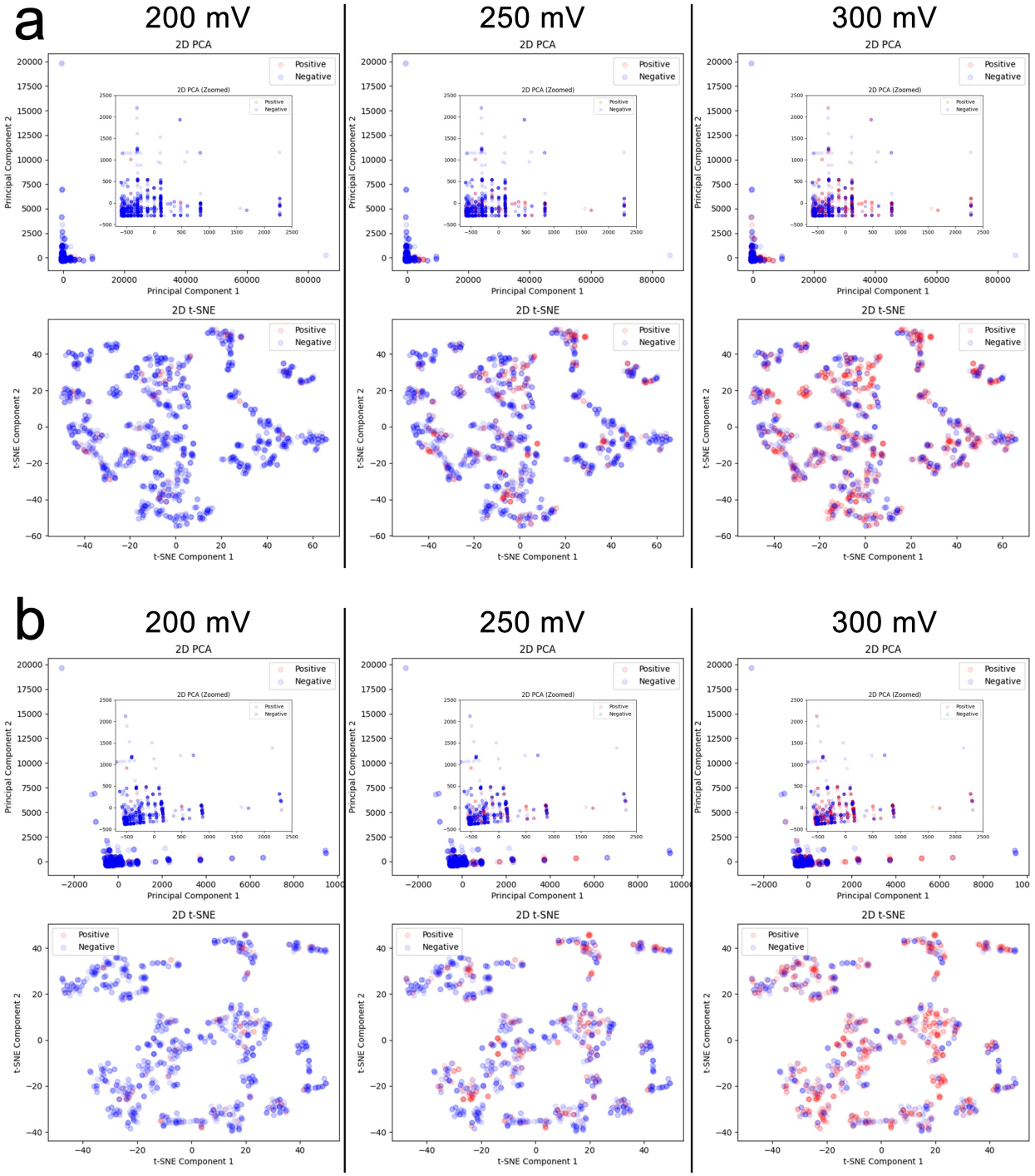

Figure SN 3-1 The two-dimensional distribution maps after PCA and t-SNE dimensional reduction transformation of the a) full and b) high-quality domain knowledge dataset, under different thresholds previously mentioned for judging the activity of OER electrocatalysts.

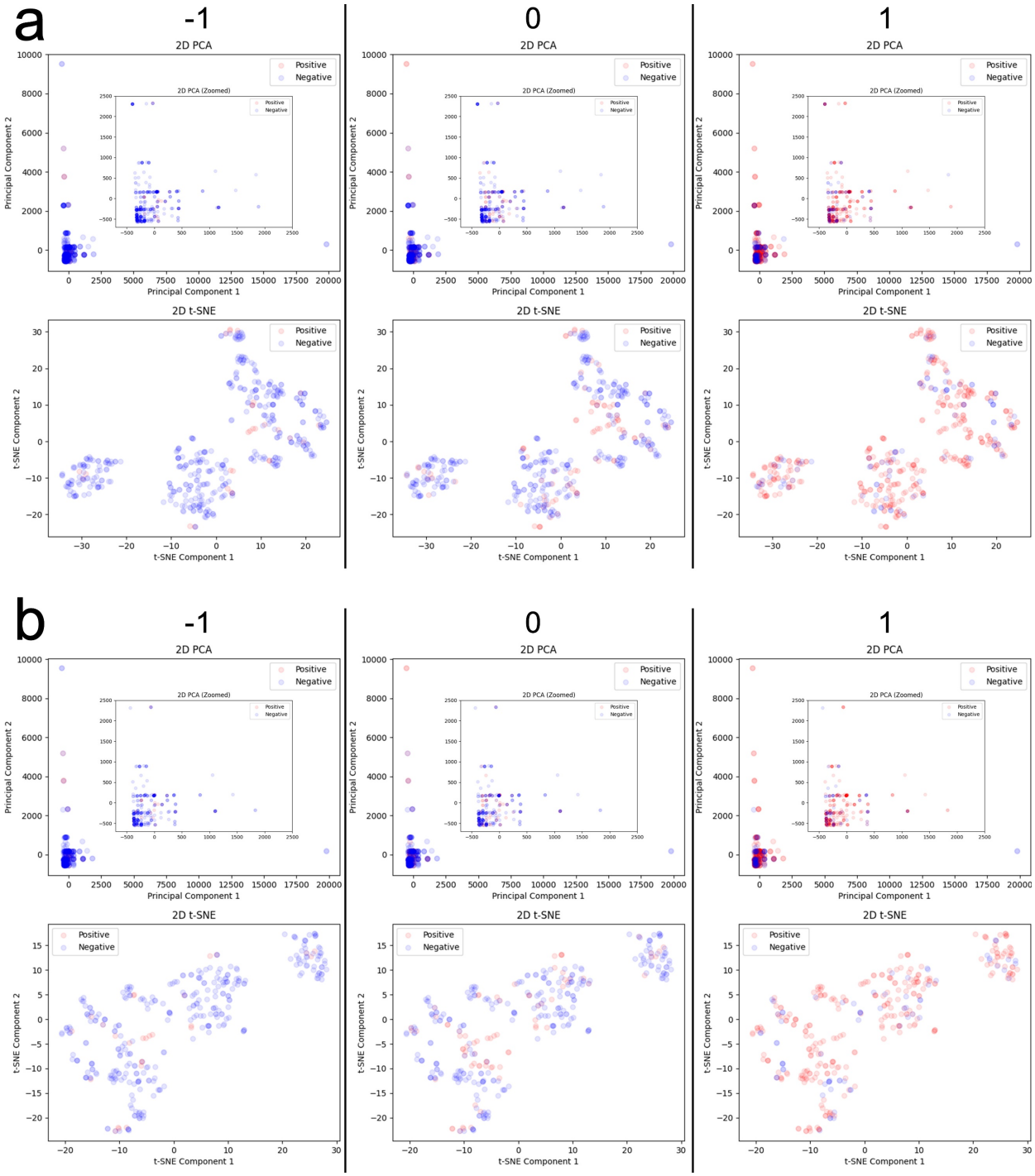

Figure SN 3-2 The two-dimensional distribution maps after PCA and t-SNE dimensional reduction transformation of the a) full and b) high-quality domain knowledge dataset, under different thresholds previously mentioned for judging the stability of OER electrocatalysts.

Table SN 3-1 Summary of the Silhouette Coefficient and Davies-Bouldin Index under different thresholds by PCA and t-SNE corresponding to **Figure SN 3-1-Figure SN 3-2**.

	Activity Full Dataset (mV)			**Activity High-Quality Dataset (mV)**			**Stability Full Dataset (mV h^{-1})**			**Stability High-Quality Dataset (mV h^{-1})**		
Thresholds	200	250	300	200	250	300	-1	0	1	-1	0	1
PCA Silhouette Coefficient	0.025	0	0.02	-0.02	0.01	0.04	0.2	0.09	0.18	0.2	0.11	0.2
PCA Davies-Bouldin Index	8.986	6.73	10.8	9.1	5.14	6.89	4.6	5.11	4	3.7	3.79	3.3
t-SNE Silhouette Coefficient	0.007	0	0	-0	0	0.01	-0	-0	0.01	-0	-0	0
t-SNE Davies-Bouldin Index	7.836	12.9	30.9	9	10.5	61	6.4	9.71	21.2	5.7	7.68	16

Supplementary Note 3 References:

1 Wold, S., Esbensen, K. & Geladi, P. PRINCIPAL COMPONENT ANALYSIS. *Chemometrics and Intelligent Laboratory Systems* **2**, 37-52 (1987). https://doi.org:10.1016/0169-7439(87)80084-9

2 van der Maaten, L. & Hinton, G. Visualizing Data using t-SNE. *Journal of Machine Learning Research* **9**, 2579-2605 (2008).